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FEDERAL SPECIFICATION

LATHE, HEAVY DUTY, FOR BRAKE DRUM AND DISC BRAKE ROTORS

This specification is approved by the Assistant Administrator, Office of Federal Supply and Services, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers one type of heavy duty lathe for turning and truing brake drums and disc brake rotors of commercial and military passenger car and truck vehicles. The lathe is electrically powered and is floor mounted with a pedestal base.

2. APPLICABLE DOCUMENTS

2.1 Government publications. The issues of the following documents, in effect on date of invitation for bids or solicitation for offers, form a part of this specification to the extent specified herein.

Federal Specifications

W-C-596	-	Connector, Electrical, Power, General Specification for
W-C-596/13	-	Connector, Plug, Electrical, General Purpose, Hospital Grade, Grounding, 2 Pole, 3 Wire, 15 Amperes, 125 Volts, 50/60 Hertz
FF-B-171	-	Bearings, Ball, Annular (General Purpose)
FF-B-185	-	Bearings, Roller, Cylindrical; and Bearings, Roller, Self-Aligning
FF-B-187	-	Bearing, Roller, Tapered
CC-M-1807	-	Motors, Alternating Current, Fractional and Integral Horsepower (500 HP and Smaller)

Federal Standards

FED-STD-H28	-	Screw-Thread Standards for Federal Services
FED-STD-123	-	Marking for Shipment (Civil Agencies)

(Activities outside the Federal Government may obtain copies of Federal specifications, standards, and commercial item descriptions as outlined under General Information in the Index of Federal Specifications, Standards and Commercial Item Descriptions. The Index, which includes cumulative bimonthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.)

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(Federal Government activities may obtain copies of Federal standardization documents and the Index of Federal Specifications, Standards, and Commercial Item Descriptions from established distribution points in their agencies.)

Military Specifications

MIL-T-152	-	Treatment, Moisture- and Fungus-Resistant. Of Communications, Electronic, and Associated Electrical Equipment
MIL-V-173	-	Varnish, Moisture-and-Fungus-Resistant (For Treatment of Communications, Electronic, and Associated Equipment)
MIL-B-13501	-	Bearings and Bushings, Brass and Bronze, Machined and Formed.
MIL-M-18058	-	Machinery, Metal and Woodworking, Packaging of

Military Standards

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-130	-	Identification Marking of US Military Property
MIL-STD-889	-	Dissimilar Metals
MS19081	-	Bearing, Roller, Tapered, Single Row of Rollers, Normal Angle, Type 751 (TS)

(Copies of military specifications and standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

Federal Regulation:

Occupational Health and Safety Administration (OSHA) 29 CFR 1910.211-222
General Industrial Safety and Health Standards

(The Code of Federal Regulations (CFR) and the Federal Register (FR) are for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402. When indicated, reprints of certain regulations may be obtained from the Federal agency responsible for issuance thereof.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply.

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American Gear Manufacturers' Association (AGMA)

AGMA 360.02 - Manual for Machine Tool Gearing
AGMA 390.03 - Gear Handbook, Volume 1, Gear Classification, Materials, and Measuring Methods for Unassembled Gears

(Application for copies should be addressed to the American Gear Manufacturers' Association, 1901 N. Fort Myer Dr., Arlington, VA 22209.)

American National Standards Institute (ANSI)

ANSI B7.1 - Safety Requirements For The Use, Care, and Protection of Abrasive Wheels

(Application for copies should be addressed to the American National Standards Institute, Inc., 1430 Broadway, New York, NY 10018.)

American Society for Testing and Materials (ASTM)

ASTM D3363 - Film Hardness by Pencil Test
ASTM D3951-82 - Standard Practice for Commercial Packaging
ASTM E18-84 - Tests for Rockwell Superficial Hardness, Metallic Materials
ASTM E380-84 - Standard for Metric Practice

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

International Organization for Standardization (ISO)

ISO 54 - Cylindrical Gears for General Engineering and for Heavy Engineering - Modules and Diametral Pitches
ISO 1122/1 - Glossary of Gear Terms - Part 1: Geometrical Definitions

(Application for copies of ISO documents should be addressed to the American National Standards Institute, Inc., 1430 Broadway, New York, NY 10018.)

National Electrical Manufacturers' Association (NEMA)

NEMA ICS 1 - General Standards for Industrial Control and Systems

(Application for copies should be addressed to the National Electrical Manufacturers' Association, 2101 L Street, NW, Washington, DC 20037.)

National Fire Protection Association (NFPA)

NFPA 70 - National Electrical Code
NFPA 79 - Electrical Standard for Metalworking Machine Tools and Plastics Machinery

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(Application for copies should be addressed to the National Fire Protection Association, Batterymarch Park, Quincy, MA 02269.)

(Industry association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 First article. When specified (see 4.3 and 6.2), the contractor shall furnish one or more complete lathes for first article inspection. The first article may be either a preproduction model or an initial production item which conforms to the requirements of this specification. In either case, the approved first article and the production items shall be identical and in accordance with the terms of the contract. Approval of the first article shall not relieve the contractor of the responsibility to furnish equipment in accordance with the requirements of this specification.

3.1.1 Contractor furnished property. Brake drums, brake discs, spindle assemblies, and bearing assemblies necessary to establish compliance with the requirements specified herein shall be furnished by the contractor. There will be no Government Furnished Property or Material (GFP/GFM) provided.

3.2 Design. The lathe shall include all the features, components, and accessories necessary for turning and truing automotive brake drums and disc brake rotors. The lathe shall be new and of the latest design capable of meeting the performance requirements specified herein. The lathe shall be electric motor driven with automatic and hand feeds. The equipment shall be convenient to operate from the front of the lathe. Operations such as feed selection, tool bar positioning, grinder-quill positioning, movement of all slides, (carriage slides, cross slides, etc.) shall be within reach of the operator. The design shall be such that parts subject to wear, breakage, or distortion shall be accessible for adjustment, repair, or replacement without modification, distortion, or damage to the lathe or any lathe part or component.

3.2.1 Measurement system. The US Customary System of Units (US) or the International System of Units (SI) may be used in the design and construction of the lathe. In this specification, all measurements, dimensions, sizes, and capacities are given in the US system. These measurements may be converted to the SI system by using the conversion factors and methods specified in ASTM E380-84.

3.2.2 Measuring and indicating devices. When specified (see 6.2), measuring and indicating devices shall be graduated in both the US and SI systems of measurement. When a dual US and SI system is furnished, dials shall have independent zero adjustments and shall be calibrated so that the last dial graduation progresses into, and is continuous with, the first dial graduation as the dial is rotated through the zero position. Dial graduations indicating stock removal or tool movement shall be in increments of not more than 0.001 inch or 0.02 mm, as applicable. When only one system of graduation is acceptable, the particular graduation required shall be as specified (see 6.2). Regardless of the measurement system used, all measuring and indicating devices on the lathe shall be graduated in the same system.

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3.2.3 Interchangeability. All parts shall be manufactured to definite standards, clearances, and tolerances in order that any such parts of a particular type or model may be replaced, interchanged, and adjusted without modification of the replacement parts or any other parts of the unit. When practical, all parts shall be permanently and legibly marked in accordance with MIL-STD-130.

3.2.4 Fastening devices. Screws, pins, bolts, and similar parts shall be installed with means for adjustment and for preventing loss of tightness. The methods for adjustment and for preventing loss of tightness shall be in accordance with accepted engineering standards and practices. All such parts when subject to removal or adjustment shall not be swaged, peened, staked, or otherwise deformed.

3.2.5 Threads. Threads shall conform to FED-STD-H28 and the applicable Detailed Standard section referenced therein.

3.2.6 Gears. All gears and pinion designs selected for use in the lathe or its components shall be manufactured to either English (USA) standard inch diametral pitch in accordance with AGMA 360.02 and AGMA 390.03 or Metric (SI) module ratio in accordance with International Organization for Standardization (ISO) ISO 54 and ISO 1121/1, and ASTM E380.84. All spindle and drive gears shall be hardened to no less than Rockwell C 42.

3.2.7 Bearings. Bearings shall be sleeve type or antifriction type. When other than tapered roller bearings are used, suitable thrust bearings shall be provided. Antifriction bearings shall be in accordance with the requirements of FF-B-171, FF-B-185, FF-B-187, or MIL-B-13501, as applicable. All bearings shall be protected from dust and chips. Bearings for the spindle and arbor support shall be tapered roller bearings.

3.2.8 Lubrication. All bearings (except sealed, permanently lubricated type), mating gears, and other moving parts which require lubrication, shall be provided means to assure adequate lubrication. Recirculating systems shall include filters which are cleanable or replaceable. Each lubricant reservoir shall have means for determining fluid levels. Oil holes, grease fittings, and filler caps shall be located to be accessible. The manufacturer shall include and install all lubricants, fluids, greases, and other compounds in accordance with the best commercial practice of the producing industry (see 6.5).

3.2.9 Safety devices. Safety devices shall be provided for parts that present safety hazards to personnel. Safety devices shall not interfere with the operation of the lathe or with the operator's performance.

3.2.9.1 Occupational safety standards. The product shall comply with OSHA standards set forth in 29 CFR 1910. 211-222 that are applicable to this type of equipment.

3.3 Material. Materials not specifically designated herein or in the contract shall be of a quality commensurate with commercial practice within the machine tool industry, shall be suitable for the intended purpose in the design of the end item, and shall meet all requirements specified herein. Materials shall be free from defects which would adversely affect the performance or

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maintainability of the individual components or the overall assembly. It shall not be permissible to use reclaimed parts as is, or rebuilt from scrap or other used equipment. When dissimilar metals are used in contact with each other, suitable protection against galvanic corrosion shall be applied in accordance with MIL-STD-889.

3.4 Construction. The lathe shall be constructed of parts and materials, which are without defects and are free of repairs. The structure shall be capable of withstanding all forces encountered during operation of the lathe at its maximum rating and capacity without distortion. Construction shall be such that, when installed and connected to the correct power supply, the lathe shall be ready for operation.

3.4.1 Castings and forgings. Castings and forgings shall be free from visible blow-holes, porosity, hard spots, cracks, scale, inclusions, cold shuts, mismatching, and other defects that could affect the structural strength of the castings or forgings. Strength and other physical and chemical properties shall be adequate to meet the performance requirements specified herein. Castings and forgings shall not be subjected to any reclaiming process without the authorization of the procuring activity.

3.4.2 Welding, brazing, and soldering. Welding, brazing, and soldering shall be of a quality which shall sustain all requirements of the welded, brazed, or soldered parts. These operations shall not be employed as repair measures for defective parts.

3.4.3 Surfaces. External surfaces shall be clean and smooth, and all edges shall be rounded or beveled. All bearing surfaces between matching parts shall be finished by machining, grinding, or scraping to necessary tolerances to provide the required accuracy. All castings, forgings, molded parts, stampings, and welded parts shall be clean and free of sand, dirt, fins, sprues, scale, flux, and other harmful materials.

3.4.4 Electrical connections. Connections to screw-type terminals shall be mechanically secured with means to prevent loss of tightness. All other electrical connections shall be mechanically secured and soldered or brazed. Only rosin base fluxes shall be used in soldering operations.

3.5 Performance and product characteristics. The lathe shall be capable of truing commercial and military passenger car and truck brake drums and disc brake rotors within the sizes specified herein, with one initial machine setup for each drum and rotor. The lathe shall be capable of truing brake drums and disc brake rotors weighing not less than 500 pounds, with the wheels removed, within the sizes and to the accuracies specified herein without the aid of an outboard support. Performance accuracy shall apply after turning or truing the brake drum or disc brake rotor in the lathe, and shall be unaffected by the mounting position of the brake drum or disc in the arbor. Performance accuracy shall also apply with the brake drum or disc brake rotor mounted on a vehicle spindle, with proper companion bearing assemblies (cups, cones and ball or roller bearings) in place and torqued in accordance with the bearing manufacturer's recommendations. Turning and truing of the drums and rotors shall be accomplished without chatter.

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If a grinder or grinding attachment (see 3.6.10.2) is required to meet the required finishes specified in paragraphs 3.5.3 (c) and 3.5.4 (c), it shall be furnished with the lathe.

3.5.1 Machine capacities, capabilities, characteristics, and requirements. Requirements shall be not less than those specified. Where a range is indicated, the required performance is from the smaller figure or less to the larger figure or more.

(a) Drum diameter range - 6 to 24 inches
(b) Drum friction surface cutting range - 1 to 9 inches
(c) Drum machine powered (not hand powered) cutting feed rate range - 0.005 to 0.015 inch per revolution
(d) Number of machine powered (not hand powered) cutting feed rates for drums - Infinite or not less than 2
(e) Disc rotor friction surface cutting diameter range capacity - 4 to 19 1/2 inches
(f) Rotor width (face to face of friction surfaces) - 0.125 to 1.75 inches
(g) Disc rotor machine powered (not hand powered) cutting feed rate range - 0.002 to 0.015 inch per revolution
(h) Number of machine powered (not hand powered) cutting feed rates for rotors - Infinite or not less than 2

For drums and disc rotors:

(i) Spindle speed range - 60 to 115 revolutions per minute (rpm)
(j) Number of machine powered spindle speeds - Infinite or not less than 3 (low speed, mid-range speed, and high speed)

3.5.2 Machine accuracy. The lathe and accessories shall not exceed the following tolerances:

(a) Spindle taper run-out (concentricity) - 0.0005 inch total indicator reading (tir)
(b) Arbor run-out (concentricity) - 0.0005 inch tir at any point along its length up to 7 inches from the spindle nose, 0.001 inch tir from 7 to 16 inches, and 0.0015 inch tir past 16 inches
(c) Adapters (such as short radii cones, taper cones, sleeves, bushings, and spacers) run-out and concentricity to their true axis - 0.0005 inch tir
(d) Carriage alignment - 0.001 inch tir with the true axis of the spindle at any point up to, and including, the necessary travel to true a nine inch face depth drum, and 0.0015 inch tir for the balance of the travel

3.5.3 Performance accuracy for brake drums. The lathe shall be capable of truing all brake drums specified herein within the following tolerances:

(a) Concentricity (turned surface to axis of hub) - the turned surface of the brake drum shall be concentric with the axis of the bearing surfaces in the hub within 0.0015 inch tir for brake drums up to 24 inches in diameter and friction surfaces up to nine inches in depth. (Applicable to rough cutting as well as finish cut.)

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(b) Taper - the turned surface of the brake drum shall be true with the axis of the bearing surfaces of the hub within 0.0015 inch tir per nine inches of brake drum friction surface for brake drums up to 24 inches in diameter. (Applicable to rough cutting as well as finish cut.)
(c) Finished surface - the final turned surface finish of the brake drum shall not have a finish numerically greater than 80 microinches arithmetical average (aa).

3.5.3.1 Rough cutting requirements for brake drums. The brake drum lathe shall rough cut brake drums of at least 24 inches diameter. The brake drum lathe shall also rough cut brake drums of at least nine inch depth. Rough cutting does not have to comply with finished requirements of para 3.5.3(c). These drums shall be rough cut as follows:

(a) At not less than 60 rpm
(b) At not less than a 0.025 inch deep cut (0.025 radius or 0.050 diameter)
(c) At not less than 0.015 inch per revolution feed rate of the carriage tool bar and cutting tool

3.5.4 Performance accuracy for disc brake rotors. The lathe shall be capable of simultaneously truing both friction surfaces of disc brake rotors, from 4 to 19 1/2 inches in diameter, within the following tolerances:

(a) Each finished trued surface of the rotor shall not deviate from the true plane of the rotor surface in excess of ±0.0005 inch or a 0.001 inch tir when measured face to face. The true plane of the rotor surface shall be a plane perpendicular to the axis of rotation which is established by the bearing surfaces of the hub (see 3.5). (Applicable to rough cutting as well as finish cut)
(b) Thickness variation (parallelism) of the rotor between trued surfaces shall not exceed 0.001 inch. (Applicable to rough cutting as well as finish cut)
(c) The finished trued rotor surfaces shall not have a surface finish numerically greater than 40 microinches aa

3.5.4.1 Rough cutting requirements for disc brake rotors. The brake disc lathe shall rough cut brake disc rotors from 4 to 19 1/2 inches in diameter:

(a) At not less than 60 rpm (medium speed)
(b) At not less than a 0.015 inch deep cut on both sides of the disc rotor, simultaneously
(c) At not less than 0.015 inches per revolution (ipr) feed rate of cross slide tool bar and cutting tool

3.6 Components. The lathe shall consist of, but not be limited to, the following principal component parts.

3.6.1 Base. The floor mounting base shall be of cast iron or of heavy steel and shall be fabricated to absorb and prevent resonant vibration and to assure strength, rigidity, and zero deformation when the brake lathe is operated under full load. Welded base structures shall be stress relieved. The base shall provide for permanent floor mounting of the lathe, with the center line of the lathe spindle at an operator working height of no less than 35 inches from the

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floor and no more than 42 inches from the floor. Compartments shall be provided in the base for the storage of short radius cones, arbors, adapters, and tools required in the resurfacing of brake drums and disc brake rotors.

3.6.2 Headstock. The headstock housing shall contain the main spindle assembly and the matching ways for the carriage. The headstock housing shall be of cast iron or cast steel construction, stress relieved, and of sufficient strength to assure rigidity and zero deformation while operating under full load. The headstock assembly shall consist of, but not limited to, the main spindle, gears, gear change mechanisms, feed change mechanisms, a handwheel drive for manually positioning the cutting tool, and the headstock ways.

3.6.2.1 Headstock ways. The headstock ways shall provide the bearing surface for tool travel (parallel to the spindle) for turning and truing brake drums. The headstock ways shall be finished to provide an accurate, uniform bearing surface for the matching carriage ways. The headstock ways shall be of sufficient length to maintain a full 100 percent load bearing contact, without any cantilever effect with the matching set of carriage ways, throughout the full carriage travel, while machining all specified sizes of brake drums. The headstock ways shall also be of adequate size and strength to maintain the required accuracies specified in 3.5.2, 3.5.3, and 3.5.4.

3.6.3 Carriage. The carriage shall provide the mating bearing surface for carriage travel (parallel to the spindle) for turning and truing brake drums. The carriage shall also provide the mating bearing surface for the cross slide travel (perpendicular to the spindle) for turning and truing disc brake rotors. The carriage shall be of cast iron or cast steel construction, stress relieved, and be of sufficient strength to assure rigidity and zero deformation while operating under heavy/full load. The carriage shall consist of, but not be limited, to the following:

(a) A handwheel drive for manually positioning the cutting tool in proximity to the work piece
(b) Provisions for locking the carriage in any position
(c) A carriage adjustment gib to compensate for wear (see 3.6.3.2)
(d) Carriage ways for the mating headstock and cross slide ways (see 3.6.3.1)

3.6.3.1 Carriage ways. The carriage ways shall be finished to provide an accurate uniform bearing surface for the matching ways. The carriage ways shall be of sufficient length to maintain a full 100 percent load bearing contact, without any cantilever effect with the matching set of ways, throughout the full carriage and cross slide travel, while machining all specified sizes of brake drums and disc brake rotors. An automatic shut off device shall be incorporated in the carriage ways that shall either disable the motor or disengage the machine powered cutting feed mechanism in the event that the ways lose the full 100 percent load bearing contact with the matching set of ways.

3.6.3.2 Carriage gib. The carriage gib shall provide for fine, precision adjustment to compensate for wear in the carriage ways throughout the life of the lathe. The carriage gib shall provide an even pressure along the surface area of the ways.

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3.6.4 Cross slide. The cross slide shall provide the matching bearing surface for cross slide travel (perpendicular to the spindle) for the operation of turning and truing disc brake rotors. The cross slide shall also provide the movable tool support for the turning and truing of brake drums and disc brake rotors. The cross slide shall be of cast iron or cast steel construction, stress relieved, and shall consist of, but not be limited to, the following:

(a) A feed dial graduated in increments to indicate 0.001 inch movement of the cross slide
(b) Provisions for locking the cross slide in any position
(c) Cross slide adjustment gib to compensate for wear (see 3.6.4.1)
(d) Provisions for mounting the tool bar in its proper position on the cross slide
(e) Cross slide ways (see 3.6.4.2)

3.6.4.1 Cross slide gib. The cross slide gib shall provide for a fine, precision adjustment to compensate for wear in the cross slide ways throughout the life of the lathe. The cross slide gib shall provide an even pressure along the surface area of the ways.

3.6.4.2 Cross slide ways. The cross slide ways shall provide the mating bearing surface for cross slide travel (perpendicular to the spindle) for turning and truing disc brake rotors. The cross slide ways shall be of sufficient length to maintain a full 100 percent load bearing contact, without any cantilever effect with the matching set of carriage ways, throughout the full cross slide travel range while machining all specified sizes of disc brake rotors.

3.6.5 Tool bar. The lathe tool bar shall be made from steel and shall be designed to hold the cutting tool and prevent change in the rake angle of the cutting tool. When specified (see 6.2), the tool bar shall be designed so that it can be rotated 180 degrees in the clamp for turning drums with external braking surfaces.

3.6.6 Machine powered feed drive. There shall be two machine powered feed systems, one for machining brake drums and the other for machining disc brake rotors. Each of the machine powered feed systems shall have either an infinite feed rate or not less than two feed rates as specified in 3.5.1, one for rough machining cuts and the other for finish machining cuts. Each machine powered feed system shall be provided with a clutch lever or a means of engaging the machine powered feeds. All feed rate changes, clutch levers, and controls shall be adjustable and accessible to the operator. All half nuts, gears, clutches, etc. required in conjunction with the machine powered feed systems shall be enclosed. Feed screws and feed rods may be exposed.

3.6.6.1 Handwheels for manual positioning. Handwheels shall be provided for manually positioning the cutting tool in proximity to the surface to be machined, to pick up the initial cut, to set the depth of cut, etc. There shall be a handwheel to provide manual movement of the carriage and another to provide manual movement of the cross slide.

3.6.6.2 Automatic stops. Automatic stop devices, for the prevention of damage to the equipment, for both grinding and turning shall be as follows:

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(a) An automatic stop device shall be provided that shall disengage the machine powered feed mechanism at a point that shall prevent any contact between the carriage and the headstock
(b) An automatic stop device shall be provided for the cross slide ways and the carriage ways, as specified in 3.6.3.1, that shall either disable the motor or disengage the machine powered feed mechanism, to prevent any overhang or cantilever effect in the event that the carriage or cross slide does not maintain the full 100 percent load bearing contact with the matching set of ways

3.6.7 Spindles and arbors. Spindles and arbors shall be alloy steel, heat treated and ground on all aligning surfaces. Spindles shall be heat treated to be surface hardness of Rockwell C 40-45. Arbors shall be heat treated to a minimum surface hardness of Rockwell C 50. Spindles and arbors shall be of a size to assure the accuracy required of this type of lathe and shall be mounted to permit turning brake drums and disc brake rotors of the size specified. The arbors shall be mounted in the spindle by means of a suitable taper (from at least 0.5 inch per foot to not more than 6.5 inches per foot on diameter) with a positive lock and release system which shall prevent damage to spindle or arbor. The length of the arbor shall be adequate to mount the sizes of disc brake rotors and hubs specified.

3.6.7.1 Spindle drive. A minimum of three spindle speeds shall be provided. Gear drives shall be totally enclosed. Adjustable tensioning devices and guards shall be provided where belt or chain drives are employed. All necessary motor mounting brackets, pulleys, belts, chains, sprockets, gears, countershafts, and other parts required to place the lathe in successful operation shall be furnished. Visual means shall be provided to show spindle speeds throughout the entire speed range specified herein.

3.6.8 Electrical equipment.

3.6.8.1 Motors. Motors shall conform to the requirements of CC-M-1807, as applicable. The lathe shall be furnished with a continuous duty, induction squirrel-cage type, minimum two horsepower motor, with a temperature rise of not more then 50°C above ambient temperature, thermal overload protected. Unless otherwise specified (see 6.2), the motor shall function on 120 volts, single phase, 60 hertz (Hz) current. Mounting shall be in accordance with NEMA ICS 1. Motors shall be protected from dirt and grit, shall be accessible for lubrication and maintenance, and shall be equipped with sealed-for-life antifriction bearings. A start-stop switch shall be mounted and wired convenient to the operator.

3.6.8.2 Electrical components. Electrical components, including relays, starters, switches, wiring, and their components shall conform to, and be located in accordance with, NEMA ICS 1 and NFPA 79, as applicable.

3.6.8.3 Light fixture. The lathe shall be provided with a work light located in a position which shall not interfere with controls but shall afford full illumination at the cutting tool area. The light shall be equipped with a separate switch that shall function independently of the motor switch.

3.6.8.4 Electric power supply cable. An electric power supply cable shall be installed on each lathe. The cable shall be not less than 10 feet in length and shall terminate with a three prong grounding plug complying with W-C-596/13

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and NFPA 70. The three prong plug shall be furnished with a 3 to 2 prong outlet adapter, with a grounding lead. The cable shall include sheathing/covering in accordance with NFPA 70 and OSHA 2206. The cable shall have a voltage/potential and amphere/current carrying capacity exceeding that required by the electric power consuming devices, including electric motors, of the lathe in accordance with NFPA 70.

3.6.8.5 Dielectric strength. The conductors and all electrical components shall withstand, without damage, twice the rated voltage plus 1000 volts root mean square (rms) at 60 Hz for a period of one minute between mutually insulated circuits, and between ground and insulated circuits.

3.6.9 Accessories. All commercial and necessary accessories shall be furnished. Ground hardened steel short radius cones shall be furnished to fit all sizes and configuration of bearing cups. For commercial and military vehicles, such accessories shall be provided to fit all bearing cups from the minimum size bearing cup MS19081-18 through the maximum size bearing cups MS19081-173 and MS19081-184 through MS19081-313. Floating tapered cone adapters shall be provided for all commercial and military brake drums and discs for commercial and military vehicles within the sizes specified in 3.5.3. Truck drum and disc anti-chatter devices which includes bands shall be provided to cover the range of sizes specified herein.

3.6.10 Optional equipment.

3.6.10.1 Case. When specified (see 6.2), a suitable case of sheet metal or wood shall be furnished to contain the grinding attachment, gages, and all accessories for transportation and storage of the equipment. The case shall be designed to prevent movement or damage to the components while in the case.

3.6.10.2 Grinding attachment. When specified (see 6.2), a grinding attachment shall be furnished with each lathe with six grinding wheels suitable for grinding cast-iron and hardened-steel brake drums. A suitable guard, an abrasive-wheel dresser having a mounted octahedron diamond of 1/4 carat minimum size, feed mechanism, brackets, and necessary equipment to make a complete unit shall be included. The grinding attachment shall be capable of driving the abrasive wheel at a working surface speed of 3,600 to 7,000 feet per minute.

3.6.10.3 Grinding wheels. Grinding wheels shall be marked, handled, mounted, and used in accordance with ANSI B7.1.

3.7 Fungus control. Components of electrical circuitry that are hermetically sealed or inherently resistant to fungus growth and moisture need not be treated. Components of electrical circuitry that are not inherently resistant to fungus growth or moisture shall be treated in accordance with MIL-T-152 and MIL-V-173.

3.8 Painting. Painting and finishing shall be in accordance with the best practices of the machine tool industry. Before painting, all surfaces shall be clean and free of all foreign matter. The paint coating shall be not less than 2.0 mil in thickness. The paint adhesion shall be such that no paint shall be peeled by fiberglass tape when two intersecting cuts expose the base metal for a distance of not less than two inches. The surface hardness of the paint shall be between the industrial standards of H to HB for pencil lead as determined in accordance with ASTM D3363.

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3.9 Product identification. The lathe shall be marked for product identification in accordance with MIL-STD-130 and, unless otherwise specified (see 6.2.1), shall include the National Stock Number.

3.10 Workmanship. Standards of workmanship shall assure that the lathe shall have the stability, strength, durability, safety, and efficient operating characteristics found in the best commercial units and as specified herein.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

(a) First article inspection (see 4.3)
(b) Quality conformance inspection (see 4.4)

4.3 First article inspection. First article inspection shall be applied to the preproduction model or initial production item which conforms to the requirements of this specification (see 3.1). First article inspection shall consist of the product examination in 4.5 and all tests under 4.6. Failure of the first article to pass the examination or any of the tests shall be cause for rejection.

4.4 Quality conformance inspection. Quality conformance inspection shall be applied to production items offered for acceptance under the contract. Quality conformance inspection shall consist of (a) through (f) below. Failure of any item to pass an examination or test shall be cause for rejection of the item.

(a) Product examination (see 4.5)
(b) Machine accuracy test (see 4.6.1)
(c) Drum truing test (see 4.6.2)
(d) Disc truing test (see 4.6.3)
(e) Dielectric strength test (see 4.6.6)
(f) Packaging inspection (see 4.7)

4.4.1 Sampling. Sampling for quality conformance inspection shall be in accordance with MIL-STD-105, using the following Inspection Levels and Acceptable Quality Levels (AQL):

Inspection or Test	Inspection Level	AQL
Product examination	II	1.0
Machine accuracy test	II	1.0
Drum truing test	S-4	1.0

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Disc truing test	S-4	1.0
Dielectric strength test	S-4	1.0
Packaging inspection	S-4	4.0

4.5 Product examination. Visually and manually examine the lathe to determine conformance with the requirements of 3.2 through 3.4.4, 3.5.1, and 3.6 through 3.10. Visual examination shall include verification of completeness of manufacture and assembly, conformance to specified standards, adequacy of markings, proper cleaning, and freedom from defects. Manual examination shall include the operation of movable parts by hand to assure proper functioning. Inspections shall be performed during the manufacturing process to the extent necessary to assure conformance; however, acceptance inspection shall be performed on completely finished (assembled) lathes. Failure of the contractor to provide objective evidence that the lathe has passed the examinations prescribed for it by the contractor's inspection system shall be cause for rejection.

4.5.1 Evidence of compliance with OSHA standards. Evidence of compliance with applicable safety and health requirements shall be reviewed (see 3.2.9.1 and 6.4).

4.6 Tests.

4.6.1 Machine accuracy test. The spindle taper run-out, arbor run-out, adapter run-out and concentricity, and carriage alignment of the lathe shall be measured. Failure to comply with the requirements of 3.5.2 shall be cause for rejection.

4.6.2 Drum truing test. Four brake drums shall be set up and turned as a typical operation. A large brake drum (minimum diameter 24 inches) and a small brake drum (maximum diameter six inches) shall be turned, as well as a wide brake drum (minimum friction surface width nine inches) and a narrow brake drum (maximum friction surface width one inch). The large brake drum shall weigh not less than 200 pounds (at the contractor's option, weights may be attached to the brake drum as necessary to meet this requirement), and shall be turned without aid of an outboard support. Performance accuracy of the lathe shall be determined by measuring the taper and concentricity of the turned surface to the spindle with dial indicators accurate within $\pm$0.0001 inch, while the freshly turned drum is still mounted on the lathe. The drum shall then be loosened on the arbor, rotated to 120 and 240 degrees from its original position, and measured for concentricity with the spindle in both positions in the same manner as specified above. For first article test only, the turned drum shall then be mounted on a vehicle spindle with proper companion bearing assemblies (cups, cones, and ball or roller bearings) in place, and torqued in accordance with the bearing manufacturer's recommendations. The turned drum shall then be remeasured for taper and concentricity with the vehicle spindle. The final finish of the turned drum shall then be checked with a surface analyzer or profilometer. Failure of the lathe to attain the performance requirements of 3.5 and the rough cut, final finish, and accuracy requirements of 3.5.3 and 3.5.3.1 shall be cause for rejection.

4.6.3 Disc truing test. Two brake discs shall be set up and treated as a typical operation. A large brake disc (maximum friction surface diameter not less than 17 inches) and a small brake disc (minimum friction surface diameter not more than five inches) shall be trued. Performance accuracy of the lathe shall be

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determined by measuring the true plane of each rotor surface and the parallelism of the rotor surfaces with dial indicators accurate within ±0.0001 inch, while the freshly turned disc is still mounted on the lathe. The disc shall be loosened on the arbor, rotated to 120 and 240 degrees from its original position, and the true planes and parallelism of the rotor surfaces remeasured in both positions in the same manner as specified above. For first article test only, the turned disc shall then be mounted on a vehicle spindle with proper companion bearing assemblies (cups, cones, and ball or roller bearings) in place, and torqued in accordance with the bearing manufacturer's recommendations. The true planes and parallelism of the rotor surfaces shall then be remeasured. The final finish of the turned disc shall then be measured with a surface analyzer or profilometer. Failure of the lathe to attain the performance requirements of 3.5 and the rough cut, final finish, and accuracy requirements of 3.5.4 and 3.5.4.1 shall be cause for rejection.

4.6.4 Spindle speed test. The lathe shall be tested with a tachometer to determine speeds and speed ranges. Failure of the spindle speeds to comply with 3.5.1(i) and 3.5.1(j) shall be cause for rejection.

4.6.5 Automatic stop test. Without a mounted brake disc or brake drum, the lathe shall be turned on and the carriage and cross-slide drives operated to the limits of their travel. Failure of the automatic stops to operate in accordance with 3.6.6.2 shall be cause for rejection.

4.6.6 Dielectric strength test. The electrical circuits shall be subjected to 1240 volts rms at 60 Hz applied between mutually insulated circuits, between insulated circuits and ground, and between each conductor and the chassis ground. Voltage shall be increased from zero to test voltage at the approximate rate of 500 volts per second, held at test voltage for one minute, and removed at the same rate as applied. During the one minute at test voltage, the circuitry shall be observed for flashover (surface discharge), sparkover (air discharge), and breakdown (puncture discharge). Any evidence of flashover, sparkover, breakdown, or damage to the unit due to heat or other effects of the applied voltage shall be cause for rejection.

4.6.7 Grinding wheel speed test. When a grinding attachment is required in the contract, the spindle speed of the grinding attachment shall be measured with a tachometer while performing grinding operations on the brake drums tested in 4.6.2. The test shall include use of each size and type of grinding wheel provided with the attachment. Failure of the attachment to maintain a calculated surface speed of 3,600 to 7,000 feet per minute for each wheel during the grinding operation shall be cause for rejection (see 3.6.10.2).

4.6.8 Paint adhesion test. A test coupon of the same material, prepared and painted at the same time and in the same manner, as the lathe frame shall be tested for paint thickness, hardness, and adhesion. For the adhesion test, two straight scratches, intersecting at right angles at their centers, shall be made in the paint with an awl, knife or other sharp instrument. Each scratch shall be not less than two inches long and deep enough to expose the base metal. A strip of fiberglass adhesive tape not less than one inch wide by four inches long shall be firmly applied over the intersection of the scratches and then removed. Peeling of any paint upon removal of the tape shall be cause for rejection. Hardness of the paint finish shall be determined by pressing the sharpened lead of a pencil into the finish with sufficient force to crumble the graphite. The

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lead of a pencil of commercial grade HB shall not penetrate the paint surface. Using the same procedure, the lead of a pencil of commercial grade H shall penetrate the surface of the paint completely to the base metal. The paint shall then be stripped from a portion of the coupon and the thickness determined by direct measurement with a micrometer accurate to ±0.001 inch, or by other means of measurement having the same accuracy as the specified micrometer. Failure of the coupon to display a coating of not less than 0.002 inch of paint shall be cause for rejection (see 3.8).

4.7 Packaging inspection. Packaging inspection shall be conducted before and after packaging to determine compliance with the requirements of Section 5.

5. PREPARATION FOR DELIVERY

5.1 Military preservation, packing, and marking. When specified (see 6.2), requirements for Level A preservation, Levels A and B packing, and military marking shall be in accordance with MIL-M-18058.

5.2 Industrial preservation, packing, and marking. Unless otherwise specified (see 6.2), preservation, packing, and marking shall be in accordance with ASTM D3951-82.

5.3 Marking for civil agencies. When specified (see 6.2), marking for shipment for civil agencies shall be in accordance with FED-STD-123.

6. NOTES

6.1 Intended use. The floor mounted lathe covered by this specification is intended to be used in military maintenance shops and depots for truing brake drums and disc brake rotors of both commercial and military passenger car and truck automotive vehicles.

6.2 Ordering data. Purchasers should select the preferred options permitted herein, and include the following information in procurement documents:

(a) Title, number, and date of this specification
(b) First article when required (see 3.1)
(c) Measuring and indicating device graduation(s) required (see 3.2.2)
(d) External tool bar rotation when required (see 3.6.5)
(e) Motor voltage, phase, and current if different (see 3.6.8.1)
(f) Case when required (see 3.6.10.1)
(g) Grinding attachment when required (see 3.6.10.2)
(h) Product identification if different (see 3.9.1)
(i) Level of preservation, packing, and marking if different (see 5.1 and 5.2)
(j) Civil agency shipment marking when required (see 5.3)

6.3 Contract data requirements. Required technical data such as operator's manuals, parts lists, and other instructions for operation and maintenance, as identified on a numbered DD Form 1664, should be specified on a DD Form 1423 incorporated in the contract.

6.3.1 Repair part manuals. Repair part identification, in repair part manuals, should include the prime contractor's name, nomenclature, part number, equipment/part manufacturer (OEM - actual part manufacturer), Federal Supply Code

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for Manufacturers (FSCM), nomenclature, part number, and price of the part immediately after the prime contractor's part identification. High mortality rate items shall be identified as shown in MIL-M-7298.

6.4 OSHA standards. Products which meet all requirements of this specification comply with the OSHA standards published prior to the date of this specification and identified in Section 3 (see 3.2.9.1).

6.5 Standard lubricants. The lathe manufacturer should specify and use to the maximum extent possible those lubricants (see 3.2.8) that are presently in the Department of Defense supply system. MIL-HDBK-113, MIL-HDBK-267, and MIL-STD-838 may be used as selection guides.

MILITARY INTERESTS:

Custodians
Army - AL
Navy - YD
Air Force - 99

Review Activities
Navy - MC
Air Force - 82

CIVIL AGENCY COORDINATING ACTIVITY:
GSA-FSS

PREPARING ACTIVITY:
Army - AL

DoD Project 4910-0604

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ACCOMPLISHMENT REPORT

Revision D of Specification OO-L-90
for
Lathe, Heavy Duty, For Brake Drum and Disc Brake Rotors

Project 4910-0604 (PECO 440-21)

The following changes have been effected by this revision:

a. Obsolete referenced documents have been deleted/replaced.

b. Material requirements have been clarified to assure use of commercial quality material, with galvanic corrosion protection required per MIL-STD-889.

c. Thread requirements were modified to permit the use of either US Customary (inch) or SI (metric) threads rather than the previously required inch threads.

d. The accuracy requirements were modified to more nearly reflect the performance expected from quality lathes produced by reputable suppliers for the commercial marketplace. To help assure a quality product, specific requirements were added for the electric power supply cable and for dielectric strength.

e. The electromagnetic interference (EMI) control requirement (and test) was eliminated as unnecessary for this type of equipment.

f. Fungus control requirements have been modified and clarified to be less restrictive for the manufacturer.

g. The option of using either the US (inch) or SI (metric) measurement system in the design and construction of the lathe was added. Additionally, the user now has the option of ordering the indicating and measuring devices in either, or both, systems.

h. The option of using industrial packaging in accordance with ASTM D3951 was added, with military (Levels A and B) protection to be provided only when so specified in the contract.

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